
```
clc;
clear;
close all;

% Battery parameters
initial_SOC = 100;      % Initial SOC (%)
capacity_Ah = 2;        % Battery capacity (Ah)
current_A = 1;          % Discharge current (A)

% Simulation time
time_hours = 0:0.1:2;

% Calculate SOC
SOC = initial_SOC - (current_A .* time_hours ./ capacity_Ah) .* 100;

% Limit SOC between 0 and 100%
SOC(SOC < 0) = 0;
SOC(SOC > 100) = 100;

% Plot SOC vs Time
figure;

plot(time_hours, SOC, 'LineWidth', 2);

grid on;

xlabel('Time (hours)');
ylabel('State of Charge (%)');

title('Battery State of Charge vs Time');

ylim([0 100]);
xlim([0 max(time_hours)]);
```

